



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD MPP.7

Math is Boundless

Lesson 10-5 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can create and describe geometric designs using repetition, patterns, and rhythm, and use a pattern unit to predict what comes next.

Language Objective: I can describe a design using the words repetition, pattern, pattern unit, and rhythm.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

repetition

pattern

pattern unit

rhythm

predictability



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The same object or shape repeated multiple times is

2

Objects or shapes repeated in a regular order form a

3

The smallest chunk of a pattern that repeats is the

4

Varying the placement, size, or order of the same shapes creates

5

Knowing what comes next because the design follows a rule is

Write About the Math

How is a design with rhythm different from a pattern?

¿En qué se diferencia un diseño con ritmo de un patrón?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **repetition**
repetición

☐ **pattern**
patrón

☐ **pattern unit**
unidad del patrón

☐ **rhythm**
ritmo

☐ **predictability**
previsibilidad

Model: The pattern unit is 3 shapes long, and $12 \div 3 = 4$ with no remainder, so shape 12 ends a complete unit — it is the blue triangle. — A strong answer divides 12 by 3, gets remainder 0, and explains that means the LAST shape in the unit. A weak answer just guesses the shape without dividing.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The circle painting's colors and sizes ____, which is why it differs from a ____ — you cannot ____ what the next circle will look like.
- My answer is ____ because ____.

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- My evidence is ____, so ____.
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** repetition
2. **Notes 2:** pattern
3. **Notes 3:** pattern unit
4. **Notes 4:** rhythm
5. **Notes 5:** predictability
6. **Watch for this mistake:** *Calling every repeated design a 'pattern.' One identical shape repeated is repetition; a pattern needs a unit of shapes repeating in a regular order; and rhythm varies the shapes' size and order, so it is neither.*
7. **Try It 1:**
8. **Try It 2:**